

AST 101
Exploring the Solar System

Prof. Lea Hirsch
Email me: lea.hirsch@utoronto.ca

Day 22
Terrestrial Planet Atmospheres

Logistics

Term Test 2 in one week - Monday, Nov 10!

Covers Ch 6, 8, 9, 10 (you do **not** need to memorize basic data about the planets!)

Topics:

- Telescopes
- Solar System patterns
- Nebular theory of planet formation
- Terrestrial planets
  - Properties
  - Geology
  - Atmospheres

Format similar to midterm 1, but a bit shorter

Review on Friday!

Term Test 1

Marks will be released at end of lecture.

Please send remark requests to my email with **“AST101 Term Test Remark”** in the subject line. I will address these as quickly as I can - but may be a week or so.

Q13 was converted to a “bonus” question.

Mean: 61.5% ; Std Dev: 18%

Learning Goals

- Describe the properties of terrestrial planet atmospheres
- Explain factors important in determining the temperatures of terrestrial planets and their atmospheric structures, including the greenhouse effect.
- Discuss mechanisms by which a planet can lose its atmosphere.
- Describe the evolutionary history of Mars’s atmosphere.

What is an atmosphere?

Layer of gas surrounding a planet:

Mostly composed of molecules like N_2, O_2, CO_2, H_2O, etc...



Terrestrial Planet Atmospheres

Moon and Mercury have essentially no atmospheres

Mars has a thin, tenuous atmosphere

Venus and Earth have “thick” atmospheres (but not compared to the giant planets)



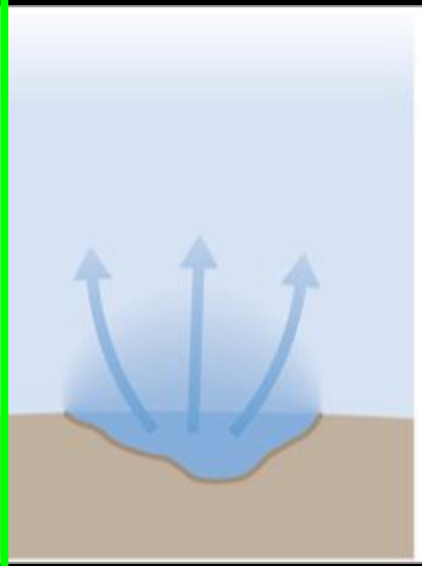
Where do atmospheres come from?

Outgassing from volcanoes

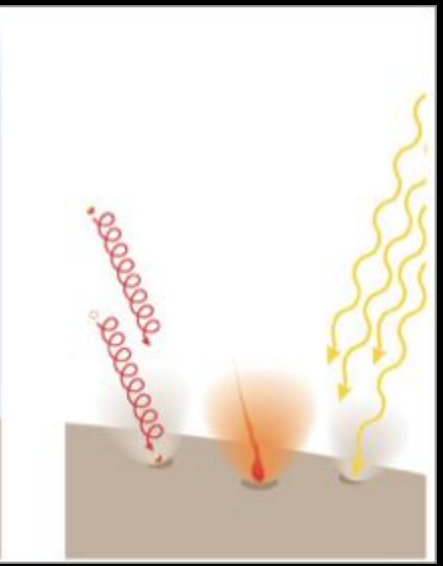
Outgassing dominates! Most of Earth/Venus/Mars atmospheres came from this source!



Evaporation



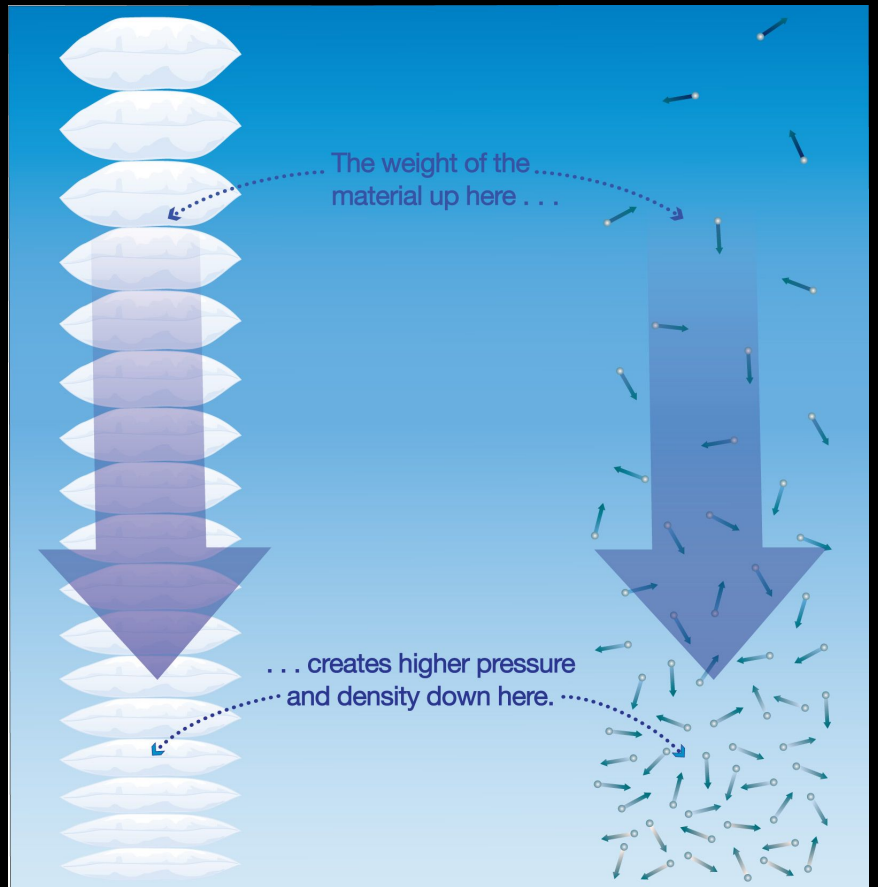
External sources - impacts of asteroids/ comets



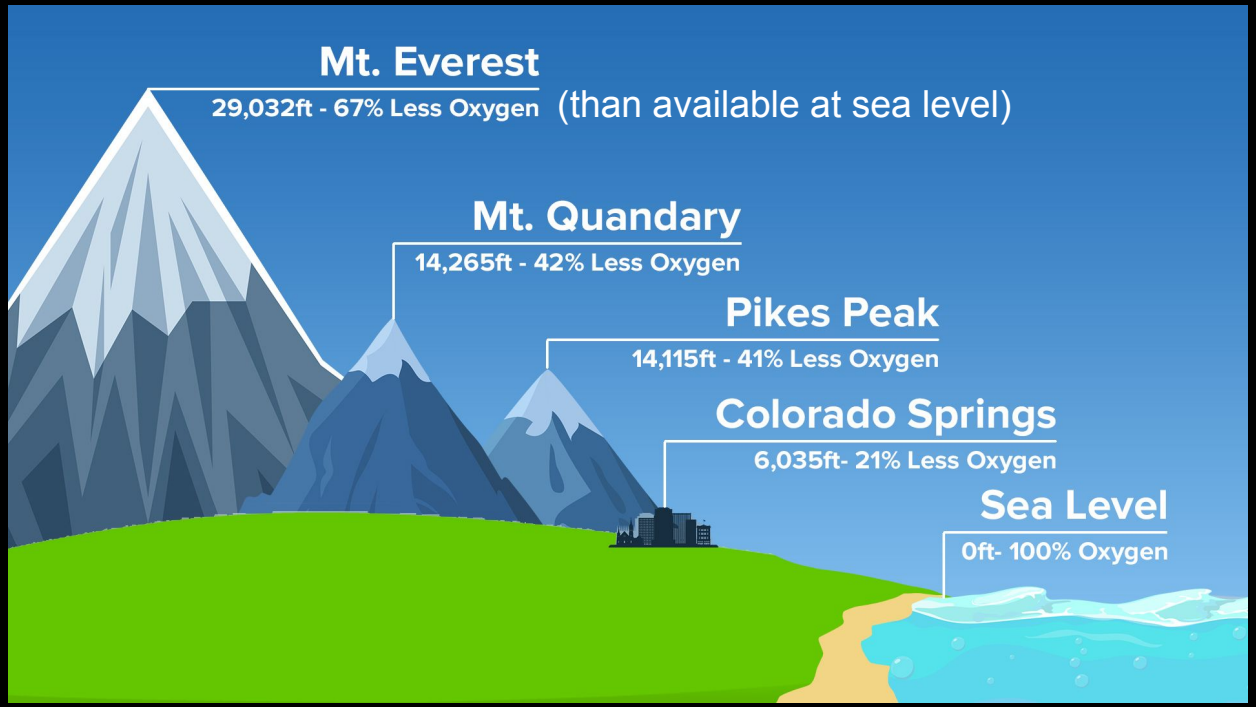
Within the atmosphere, collisions between molecules creates pressure.

Higher densities and higher temperatures cause higher pressures.

- Higher density = more particles to collide with each other
- Higher temperature = particles are moving faster



Pressure and density of the air declines with height above sea level.



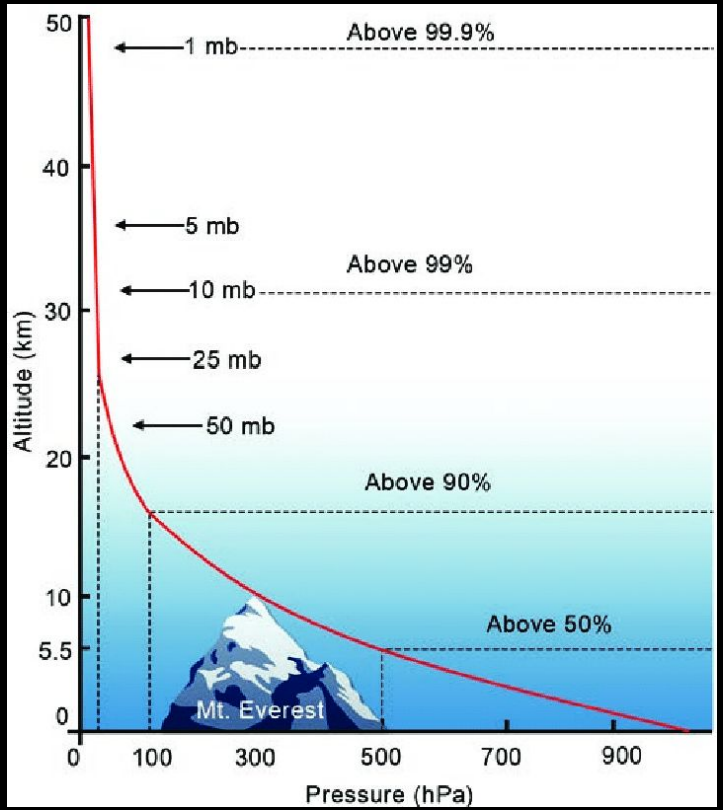
Discuss with a neighbor:

How would you define the top of the Earth's atmosphere?

Use the graph to the right to make an argument for where the top of the Earth's atmosphere is.

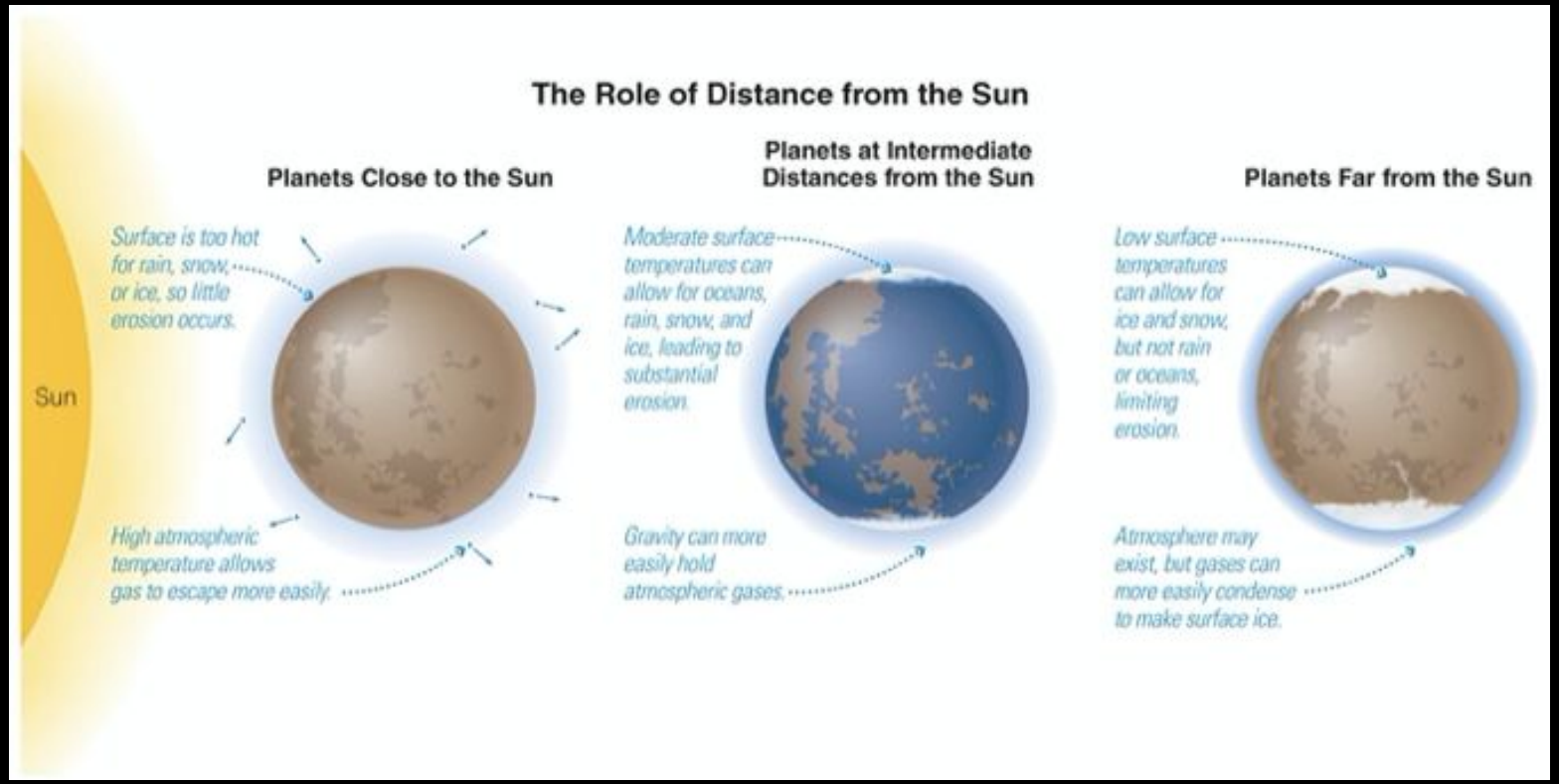
What fraction of Earth's radius (~6000 km) does the atmosphere represent?

Note: 1000 hPa (hectopascal) = 1000 mb (milibar) is the pressure at Earth's surface.



What Determines the Temperature on the Surface of a Planet?

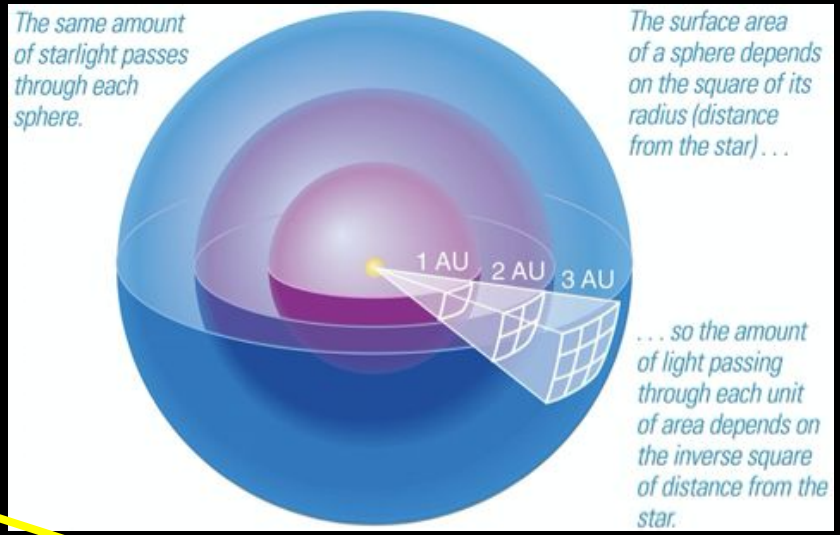
Orbital Position vs. Planet Conditions



How much energy is received from the Sun?

The farther you go from the Sun, the more its light is spread out.

The power absorbed by each square meter of a planet depends on its distance from the Sun.



$$\text{Flux received} = \frac{L_\odot}{4\pi d^2}$$

Flux [Watts / m^2]

Luminosity (power output) of the Sun [Watts]

Distance from the Sun [m]

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If the Sun's luminosity increased by a factor of two, which of the following would be true?

$$\text{Flux received} = \frac{L_\odot}{4\pi d^2}$$

- Earth would receive 4x more flux (energy per second per square meter).
- Earth would receive 2x more flux.
- Earth would receive the same flux.
- Earth would receive 1/2 as much flux.
- Earth would receive 1/4 as much flux.

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Determining a Planet's Temperature

Without an atmosphere, the surface temperature of a planet would depend on:

1. How far it is from the sun (how much **flux** it receives).
2. How reflective its surface is (how much of that flux is absorbed vs. reflected back to space)
    - **Reflectivity (Albedo):** ranges from 0 (fully absorptive) to 1 (fully reflective)

$$T_{\text{“no greenhouse”}} = 280\ K \times \left(\frac{1 - reflectivity}{d^2} \right)^{1/4}$$

AU

Note: This equation only works for planets in the solar system, since it doesn't account for potential variations in stellar luminosity!

Let’s try using the equation

Calculate the approximate temperature of one of the following planets:

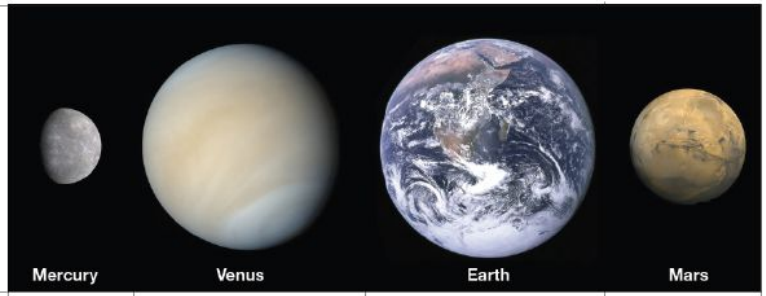
$$T_{\text{“}no\ greenhouse\text{”}} = 280\ K \times \left(\frac{1-reflectivity}{d^2}\right)^{1/4}$$

Venus: d = 0.723 AU, reflectivity = 0.75

Earth: d = 1.000 AU, reflectivity = 0.30

Mars: d = 1.523 AU, reflectivity = 0.25

We can use this equation to predict the temperatures of the terrestrial planets with no knowledge of their atmospheres:

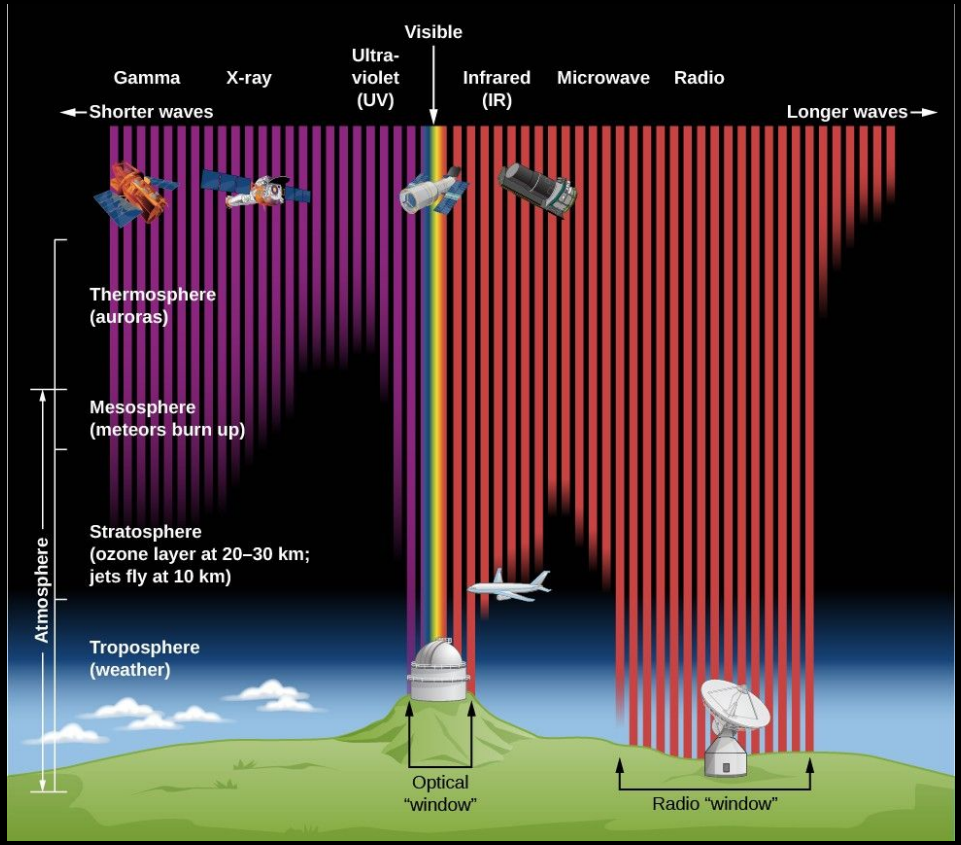
Planet				
Distance from sun (millions of km)	57.9	108.2	149.6	227.9
Insolation (W/m^2)	2290	662	342	145
Albedo	0.10	0.75	0.30	0.25
$T_{predicted}$ (°C)	163	-41	-18	-64



Atmospheres Absorb and Scatter Incoming Light

This can be inhibitive (like when we want to do astronomy from the ground).

But it can also be helpful (like when solar UV rays are blocked out).

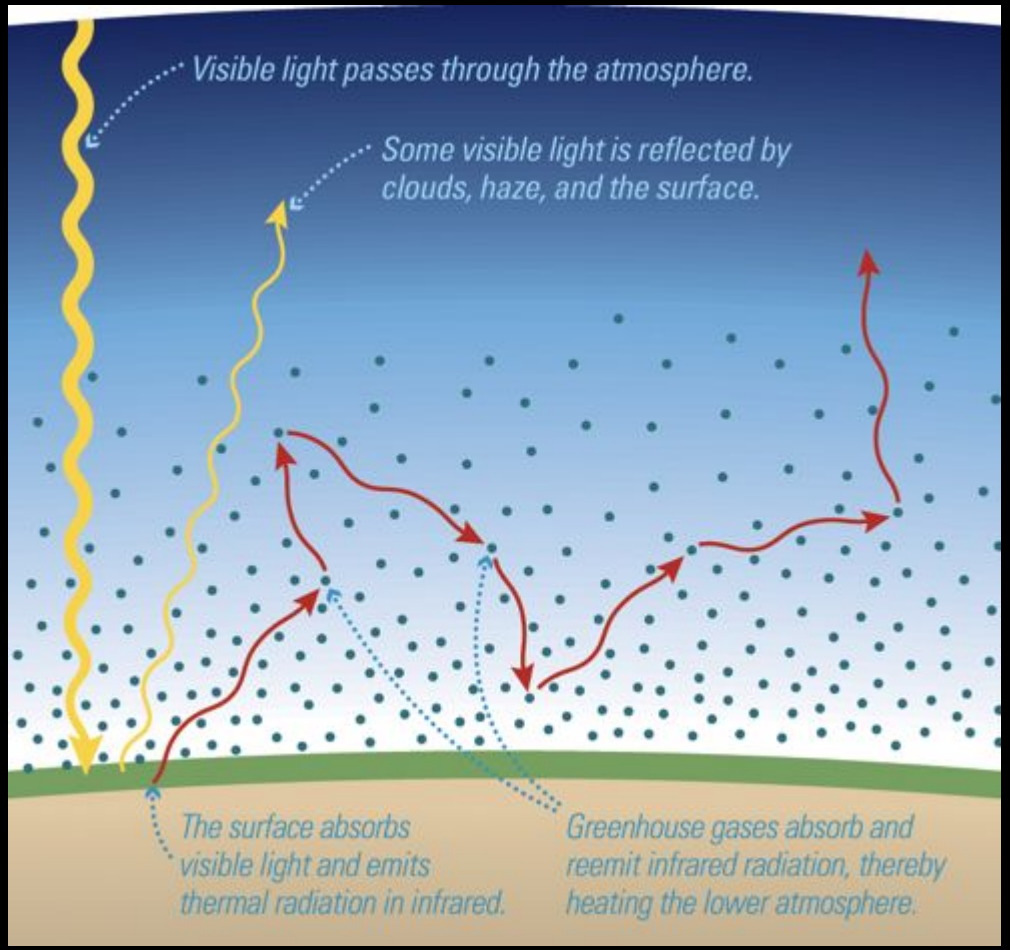


Greenhouse Effect

Visible light passes through the atmosphere and heats the surface.

The surface emits thermal (blackbody) radiation, primarily in the infrared.

Greenhouse gases in the atmosphere absorb infrared light, trapping heat near the surface.



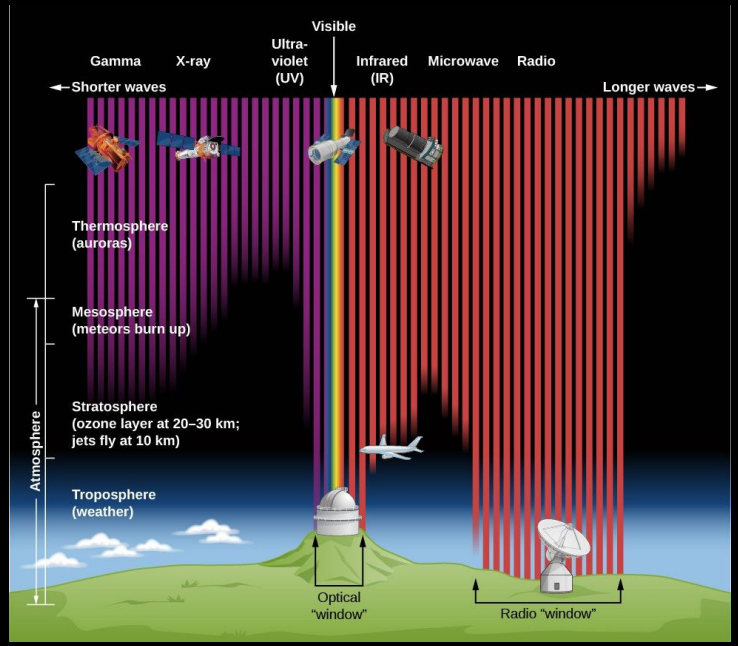
Which of the following are significant greenhouse gases in Earth's atmosphere?

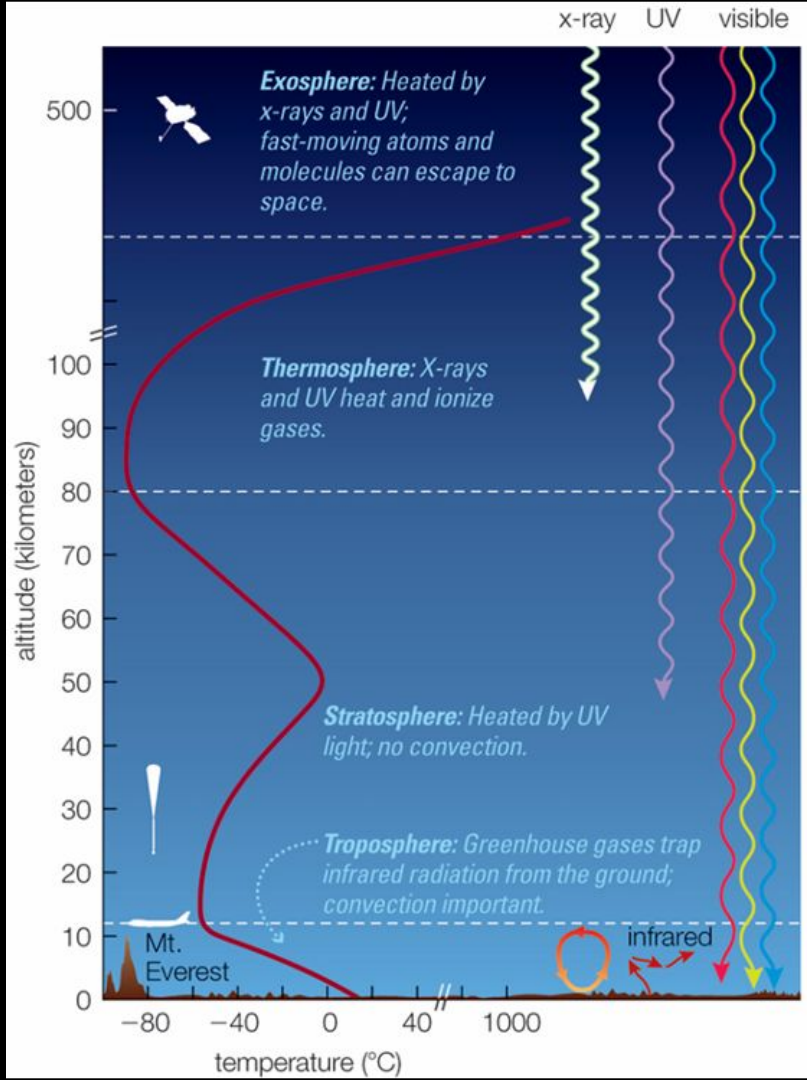
A. Carbon dioxide CO_2
B. Water vapor H_2O
C. Methane CH_4
D. Nitrogen N_2
E. Oxygen O_2

Greenhouse gases: absorb infrared photons

Major atmospheric constituents, but do not absorb infrared photons.

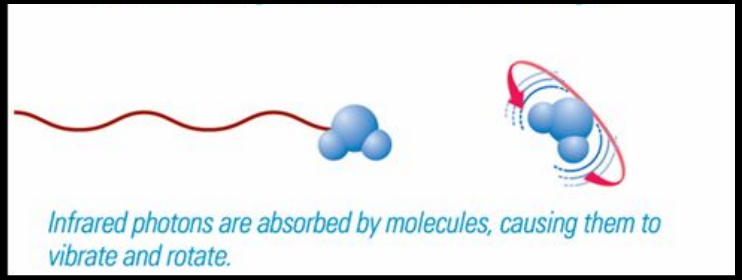
Atmospheric Layers



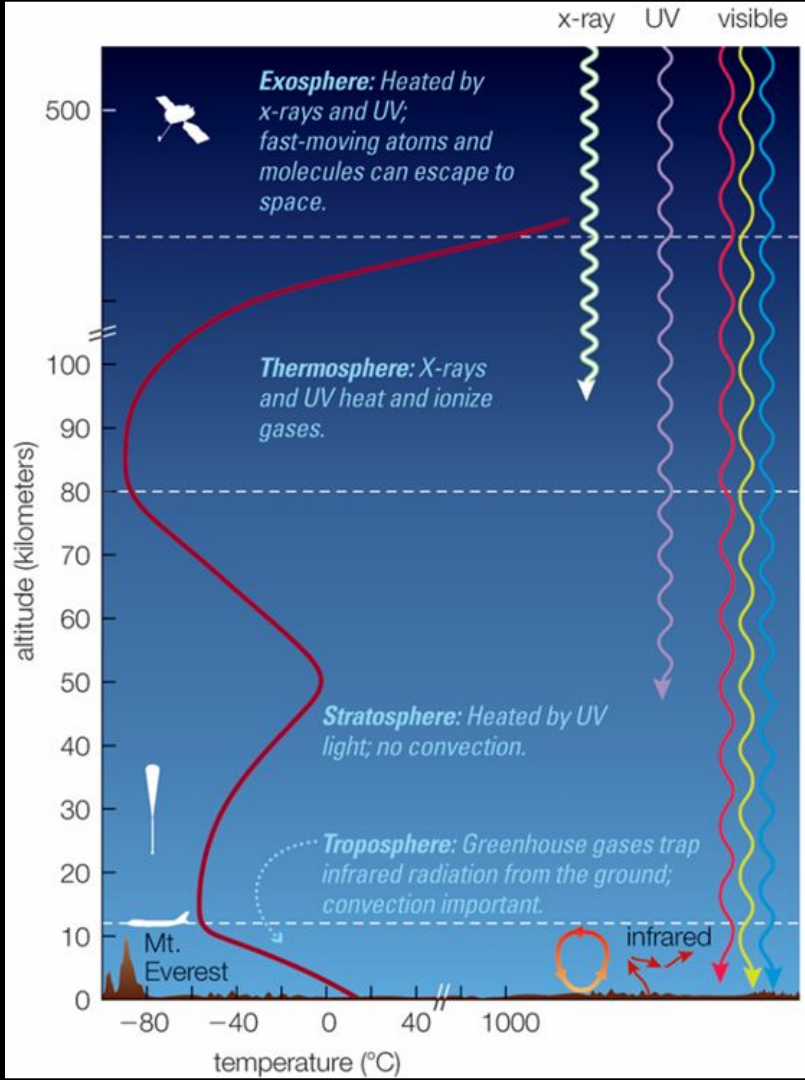


Interactions between photons and atmospheric molecules:

IR photons emitted by Earth's surface are absorbed by greenhouse gas molecules. This heats the bottom of the troposphere.

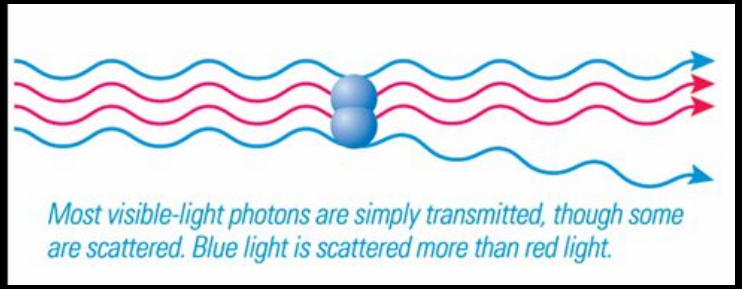


Infrared photons are absorbed by molecules, causing them to vibrate and rotate.

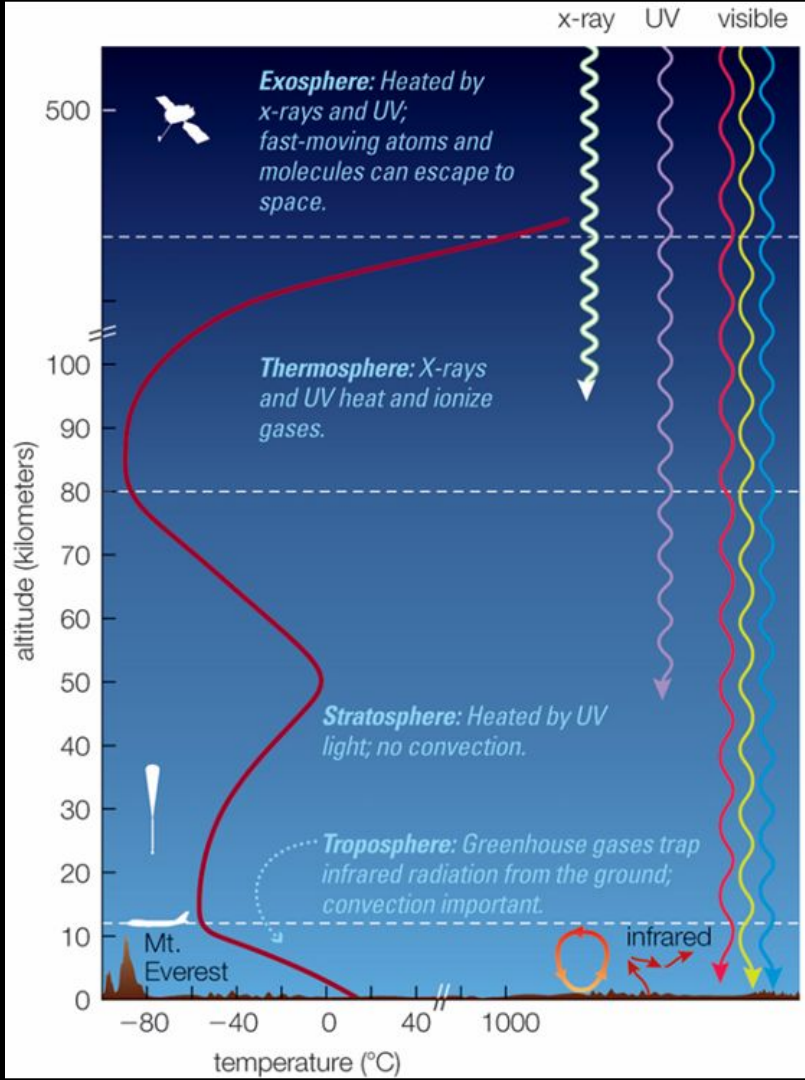


Interactions between photons and atmospheric molecules:

Visible light passes through, but blue photons are scattered more than red. This light heats up Earth's surface.

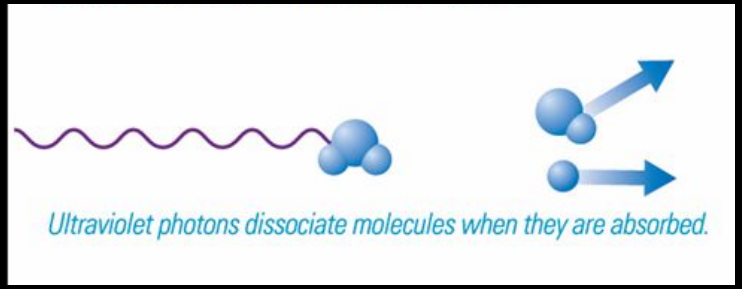


Most visible-light photons are simply transmitted, though some are scattered. Blue light is scattered more than red light.

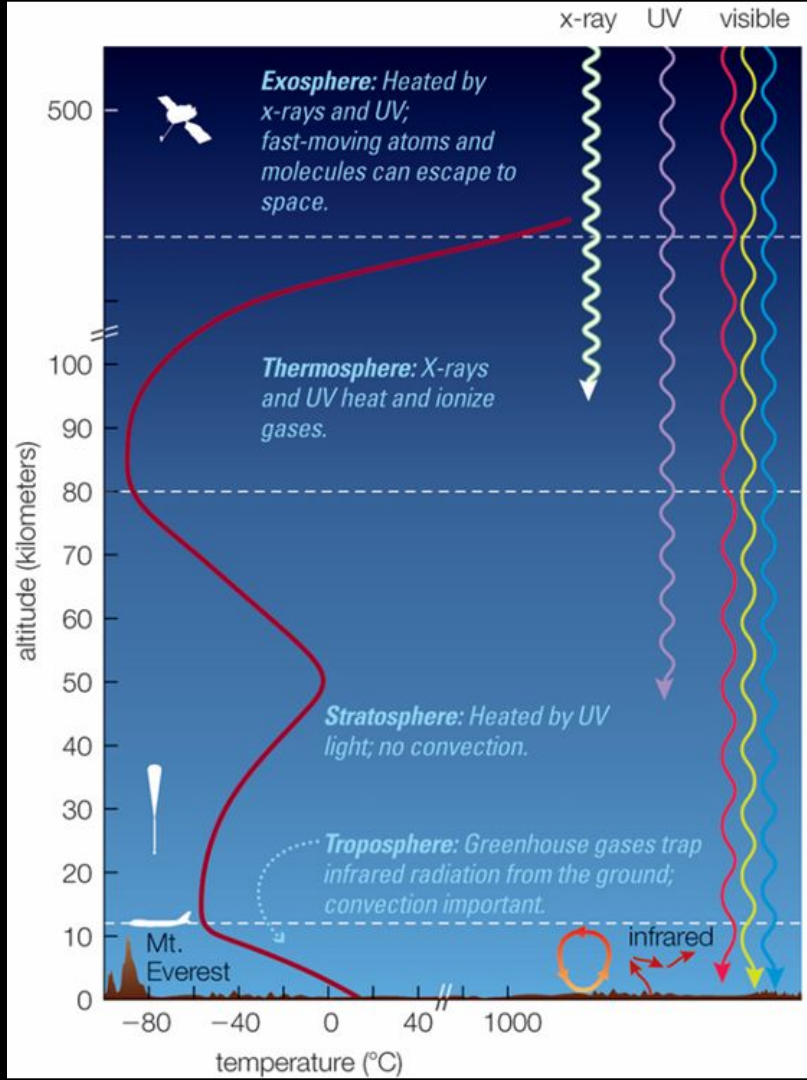


Interactions between photons and atmospheric molecules:

Ultraviolet photons **dissociate** molecules (e.g. O_3) in the stratosphere / ozone layer.

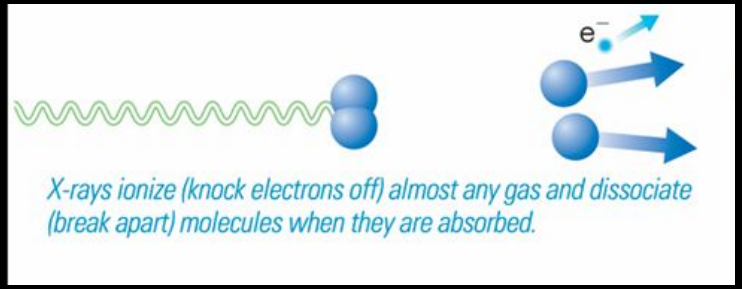


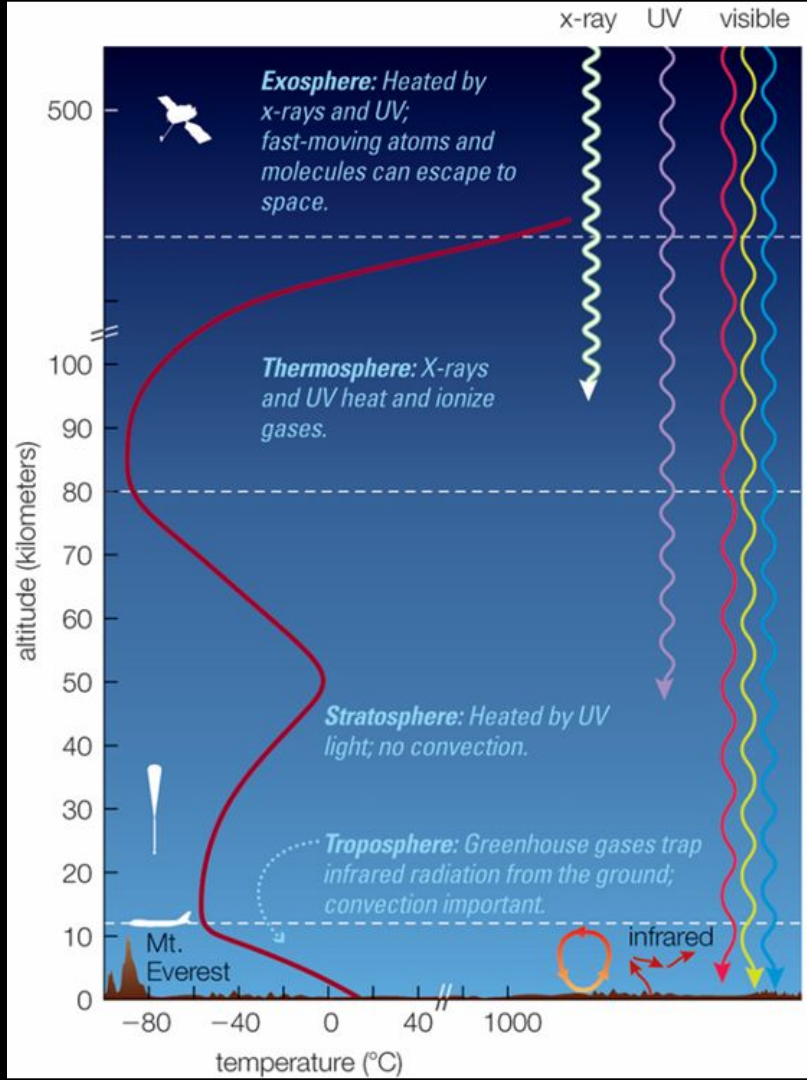
Ultraviolet photons dissociate molecules when they are absorbed.



Interactions between photons and atmospheric molecules:

High-energy photons (x-rays, far UV) **dissociate** molecules and **ionize** gas in the exosphere and thermosphere:

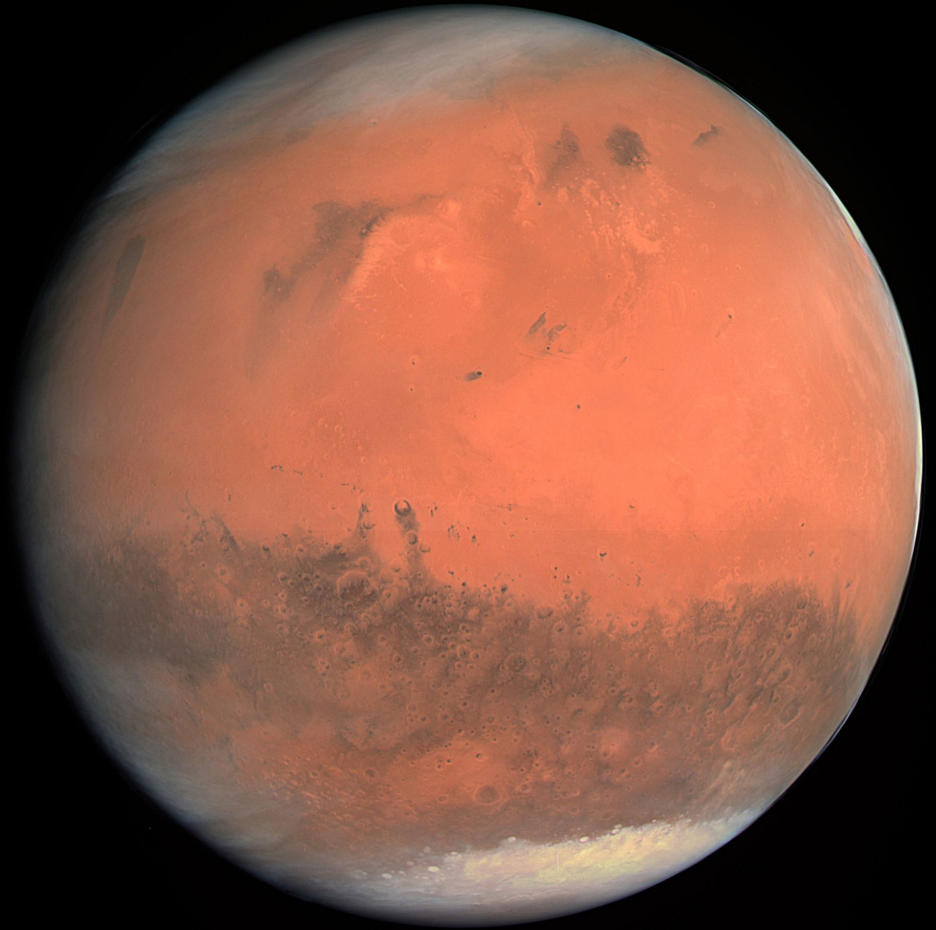




Mars

Why is Mars's atmosphere so thin and tenuous?

Why is Mars's surface too cold for liquid water?



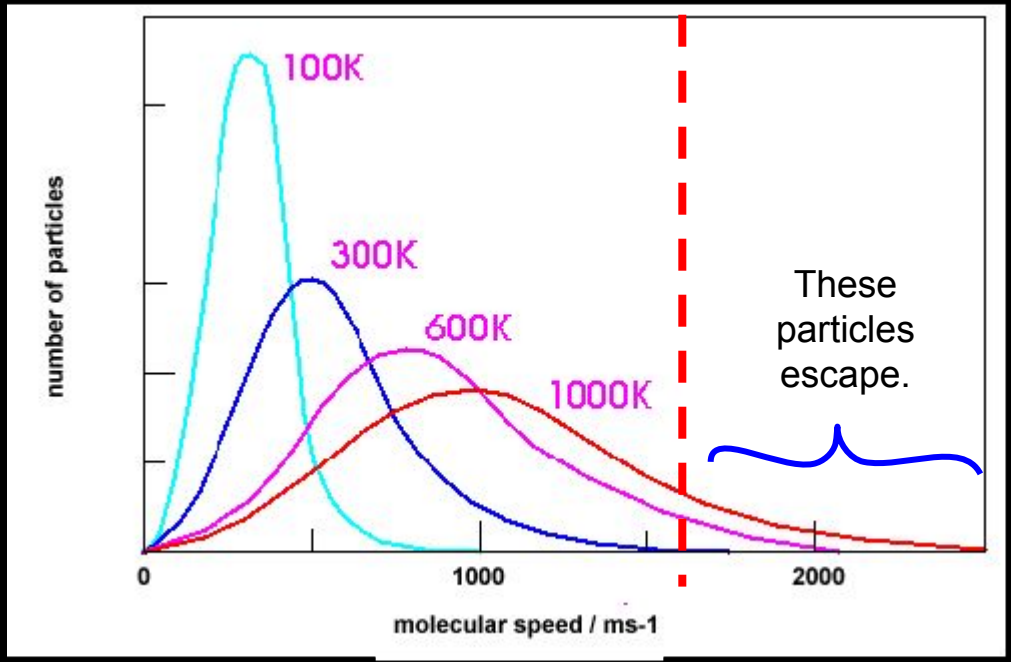
How can a planet lose its atmosphere?

$$v_{esc} = \sqrt{\frac{2GM}{R}}$$

Thermal Escape:

Temperature is a measure of the average kinetic energies of the particles.

At a given temperature, the number of particles at a given speeds follows a distribution as shown:



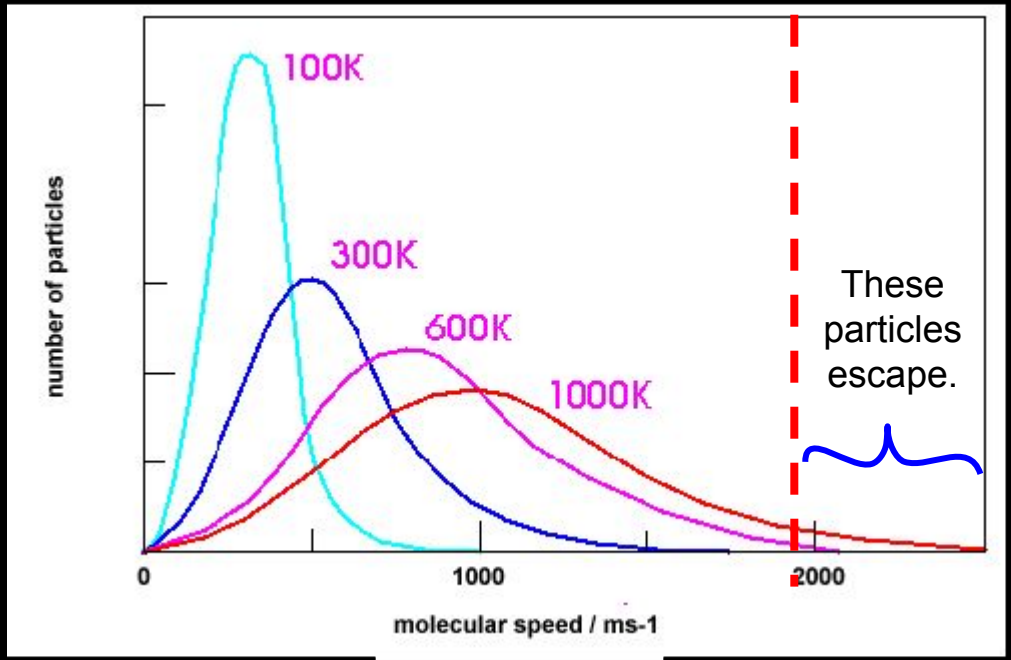
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All else being equal, which planet would lose its atmosphere faster?

A low-mass planet

A high-mass planet

Mass is irrelevant to atmospheric loss.

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All else being equal, which planet would lose its atmosphere faster?

A hot planet

A cold planet

Temperature is irrelevant to atmospheric loss.

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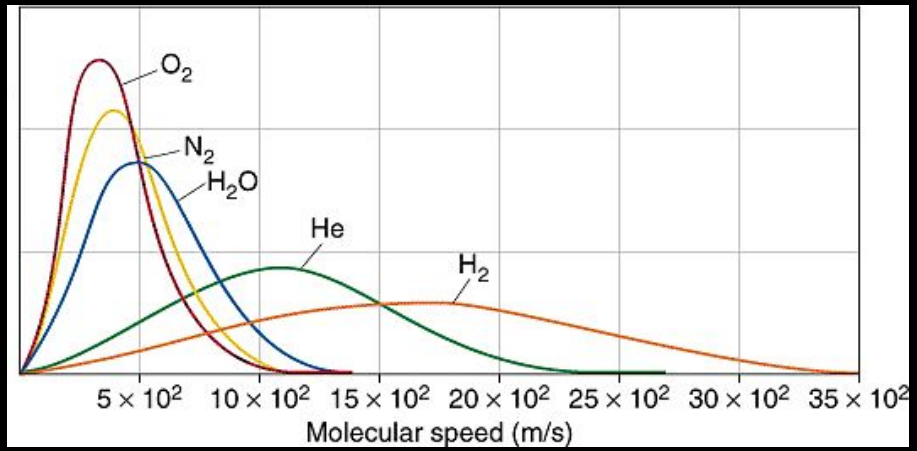
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Thermal Escape

Happens fastest for planets with:

- Low masses (low escape velocities)
- Strong irradiation (high temperatures)

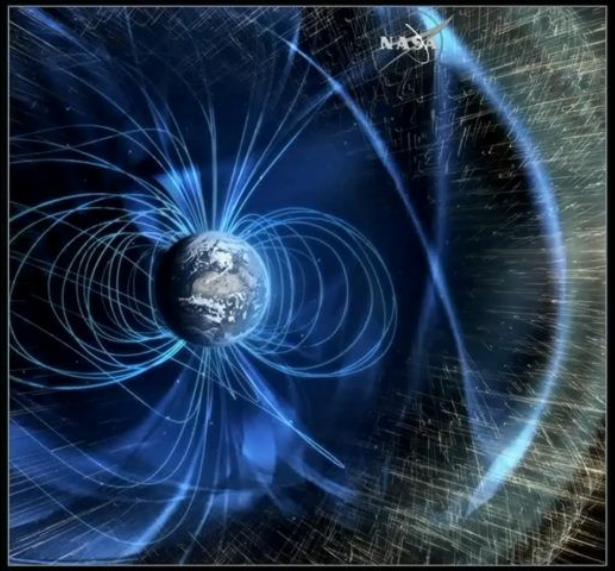
Happens fastest for the lightest elements (e.g. hydrogen) because they move faster at the same temperature.

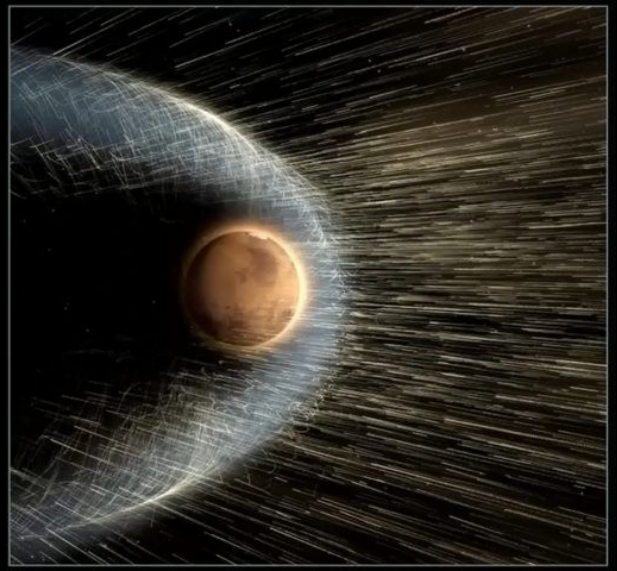


How can a planet lose its atmosphere?

Solar Wind of charged particles (electrons and ions)

Earth’s magnetic field protects its atmosphere from charged particles by deflecting them around the magnetosphere.

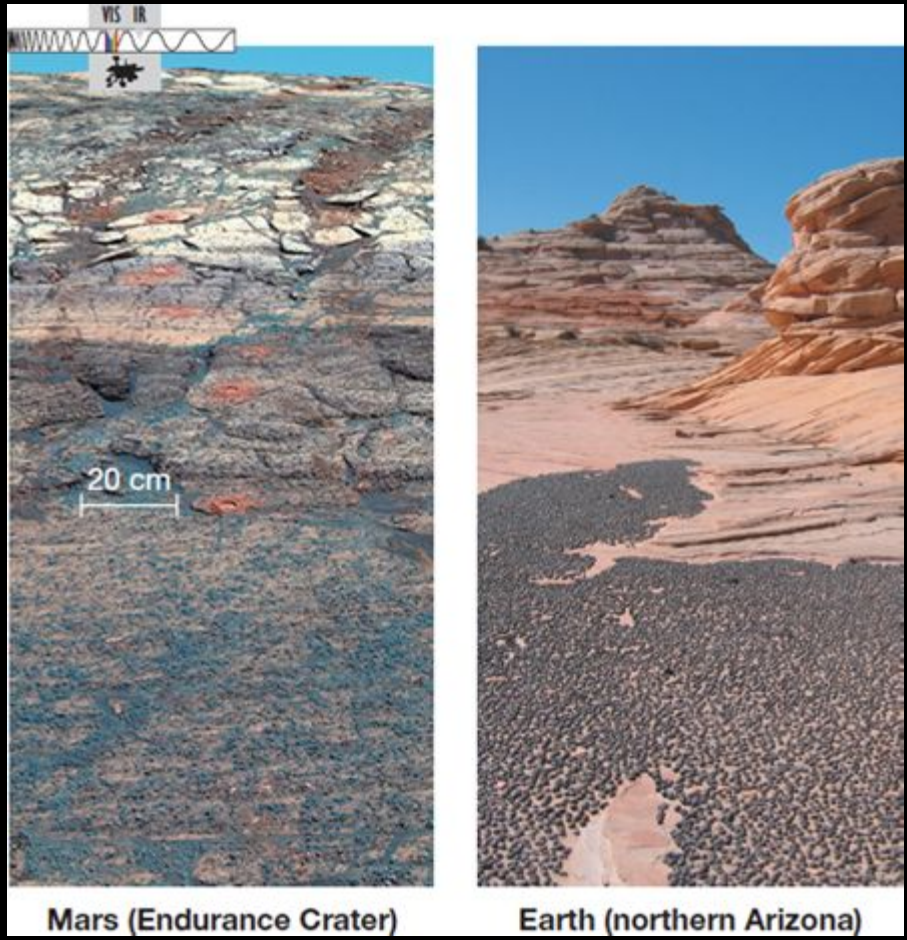




Atmosphere of Mars

Liquid water used to be abundant on Mars.

“Blueberries” - round pebbles on Mars resemble rounded pebbles formed in aqueous environments on Earth



Atmosphere of Mars

Liquid water used to be abundant on Mars.

Other rock types also resemble rocks from streambeds on Earth.





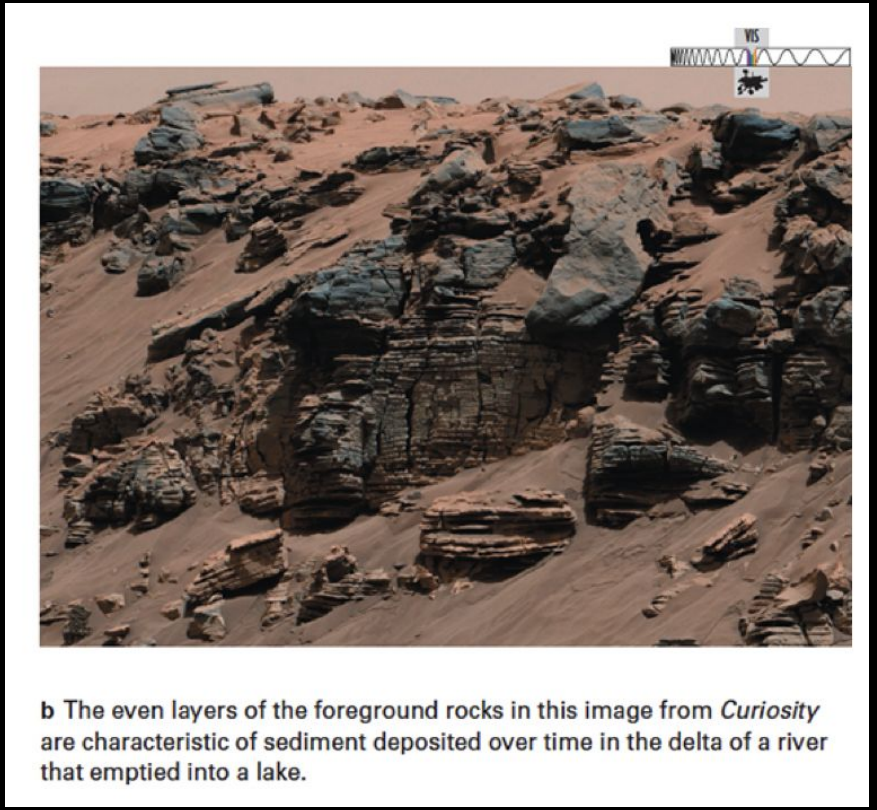
Mars Earth

a Clumps of rounded pebbles found by *Curiosity* in an ancient streambed on Mars show a structure nearly identical to that found in a typical streambed on Earth, providing strong evidence that the pebbles were rounded by flowing water.

Atmosphere of Mars

Liquid water used to be abundant on Mars.

Layering of rocks indicates sedimentary deposits in an aqueous environment.

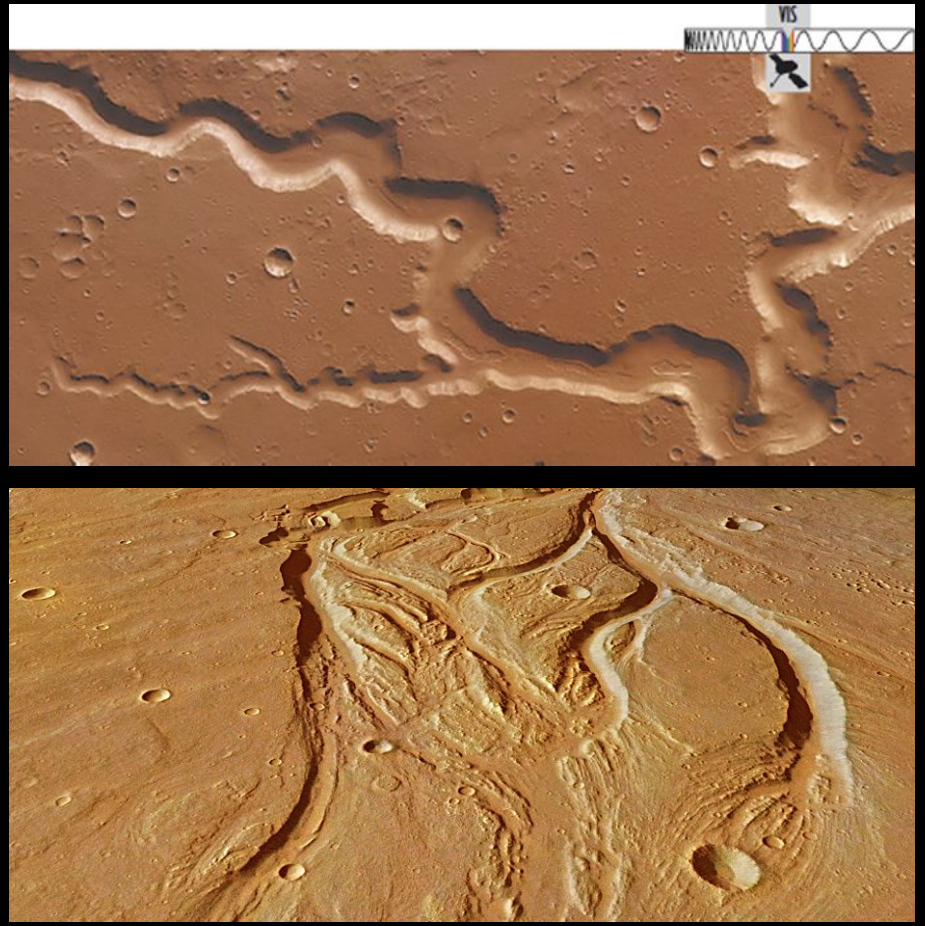


b The even layers of the foreground rocks in this image from *Curiosity* are characteristic of sediment deposited over time in the delta of a river that emptied into a lake.

Atmosphere of Mars

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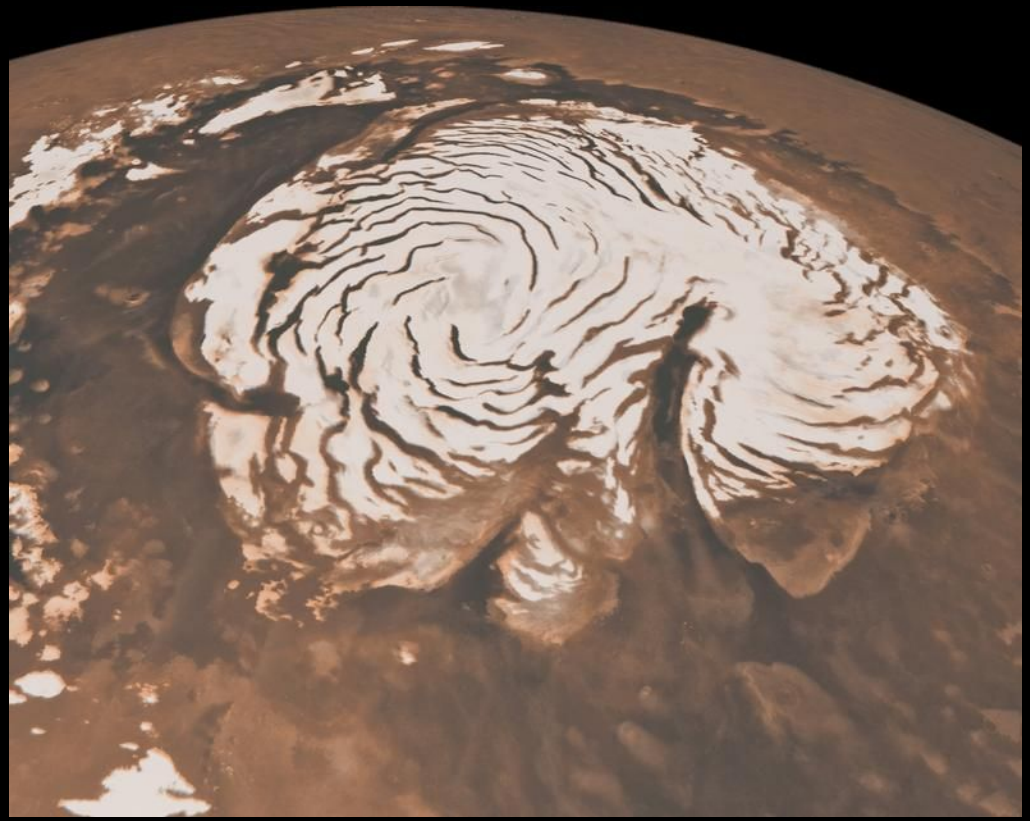
Large-scale topography resembles dry riverbeds and river deltas.



Currently, Mars's water is frozen in its ice caps and underground.



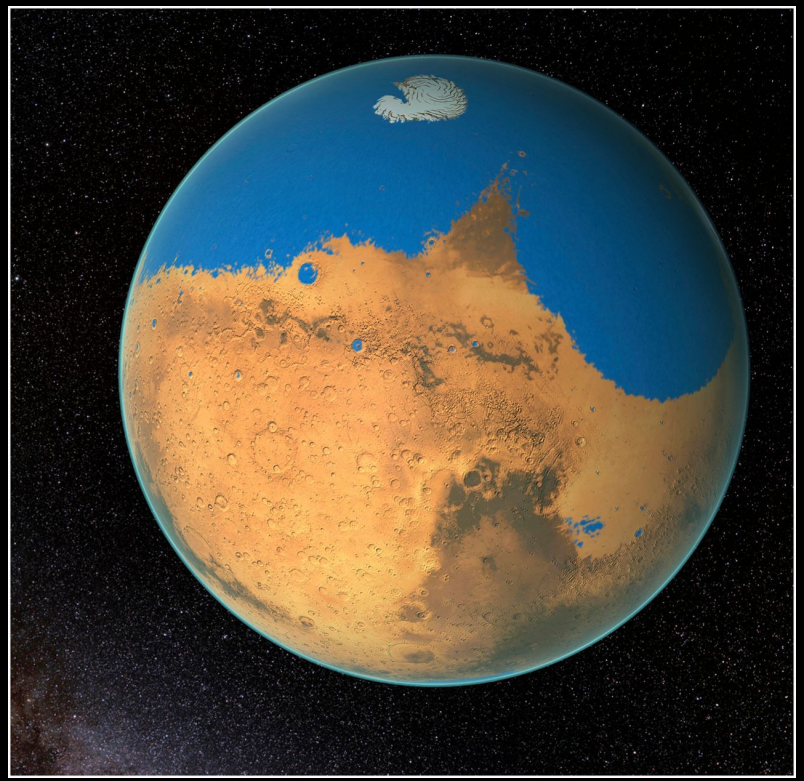
Landslide caused by weakening of layers of icy material in slightly warmer conditions. (MRO 2010)



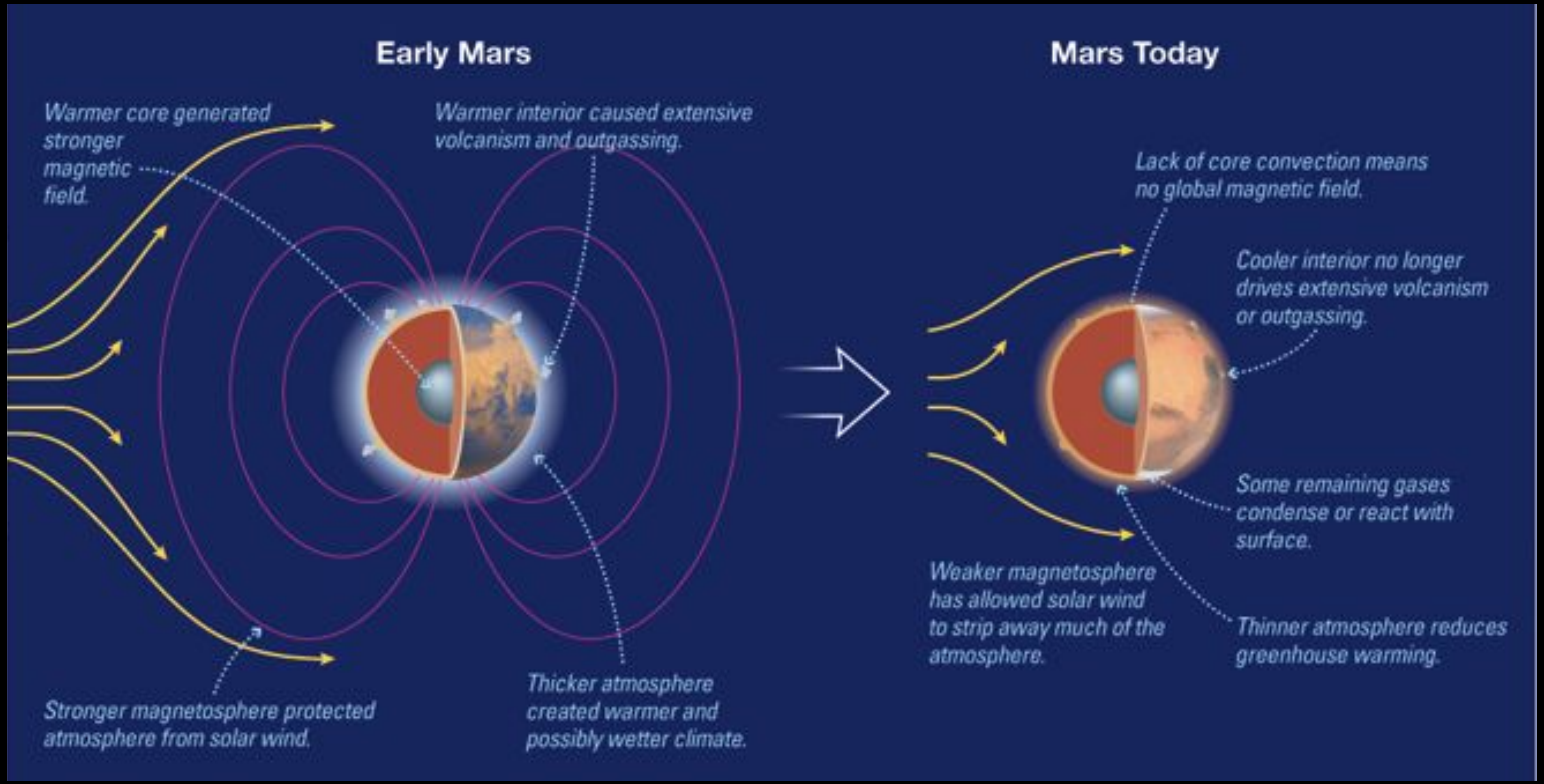
Liquid surface water would have required higher temperatures, perhaps supplied by a greenhouse effect.

Could Mars have had a giant ocean and a much thicker atmosphere?

(Artist’s impression)

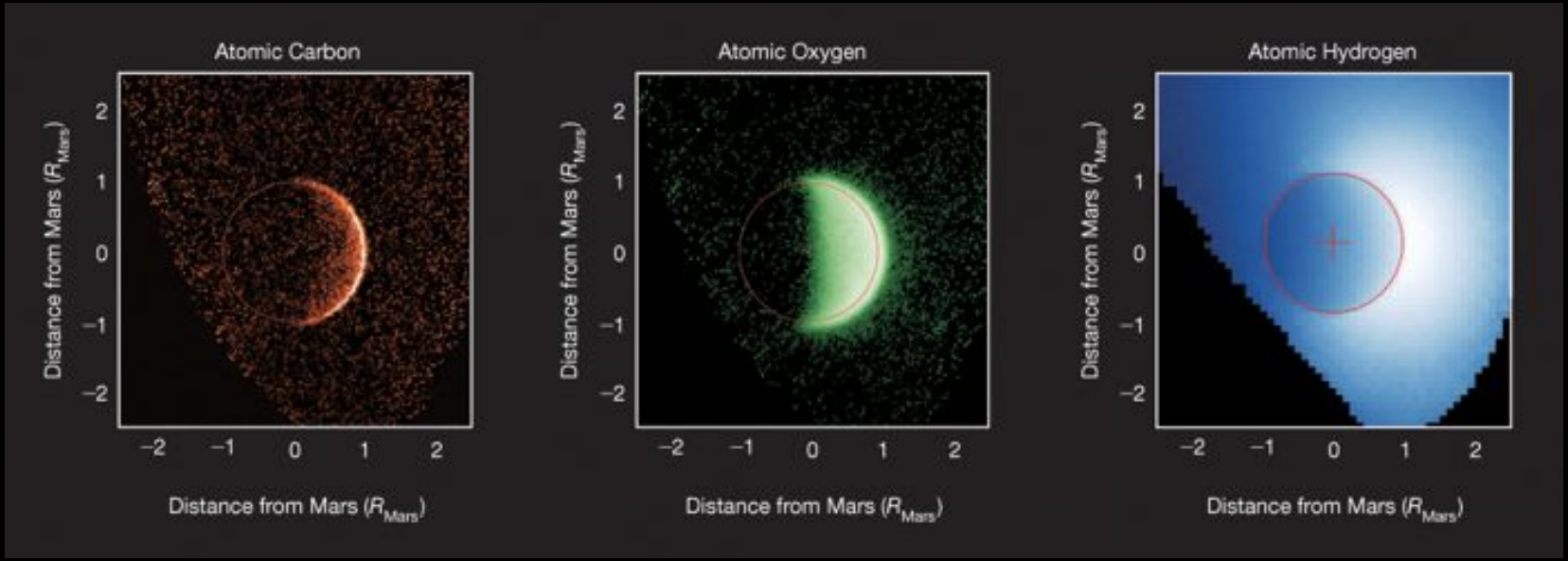


As Mars cooled down and convection stopped, it lost its magnetic field, and therefore its protection from the solar wind.



Atmospheric Loss is ongoing and measurable.

Carbon, oxygen, and hydrogen are all escaping Mars via solar wind (C, O) and thermal escape (H).



Discuss:

Which of the following attributes of Mars was most responsible for its atmospheric and climatic evolution?

- Mars’s small size and low mass (compared to Earth)
- Mars’s distance from the Sun
- Mars’s rotation rate